

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	Slot B	Date:	17.6.26

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker or graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000). Analyse how changing a hexadecimal value affects the binary representation and the displayed output with suitable illustration.
2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry, and Parity flags after each operation. Analyse how the flag register influences program execution and determines why different inputs produce different flag values.
3. A user notices that a computer responds slowly while editing a document, browsing the internet, and playing audio simultaneously. To analyse the system behaviour, perform these activities together and observe the system's response. Analyse how the CPU, memory, and I/O devices cooperate to execute these tasks, and identify which functional unit is primarily responsible for processing, storing, and receiving data during the operations.

Code of Conduct (192512274)

I R. Madhumitha R (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R. Madhumitha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge



UPGRADING SOFTWARE

(ARM Cortex, Core i5 or higher)



1. WHY UPGRADE?

- To use modern architectures that offer better instruction execution.
- To improve cache utilization, multi-tasking capability and overall performance.



Benefits

- ✓ Faster execution
- ✓ Better performance
- ✓ Efficient cache usage
- ✓ Support for modern applications
- ✓ Improved user experience
- ✓ Future ready technology

2. KEY FEATURES OF MODERN ARCHITECTURES

- Pipelining
- Out-of-Order Execution
- Multi-core & Multi-threading
- Larger Cache Memory
- SIMD Instructions
- Power Efficiency
- Advanced Security Features

3. HOW UPGRADING INFLUENCES PERFORMANCE

Performance Improvement



Instruction Execution



More instructions executed per clock cycle (IPC).

Cache Utilization



Larger and smarter caches reduce memory access delay.

Multi-tasking



More cores and threads allow parallel processing.

Overall System Performance



Higher throughput, less latency and better efficiency.

4. MODERN APPLICATION AREAS BENEFITED

- Artificial Intelligence
- Machine Learning
- Data Science
- Cloud Computing
- Multimedia & Gaming
- Real-time Systems

5. STEPS FOR UPGRADING

Step 1



Assess current system requirements.

Step 2



Choose modern architecture (ARM Cortex / Core i5 or higher).

Step 3



Update hardware (if required) and install new software.

Step 4



Optimize settings and drivers.

Step 5



Test, benchmark and validate performance.

CONCLUSION



Upgrading to modern architectures like ARM Cortex or Core i5 (or higher) improves instruction execution, cache utilization, multi-tasking and overall system performance, enabling efficient handling of modern applications and future technologies.



Computer Memory (RAM and Cache)

1. What is Computer Memory?

A storage area used by the CPU to store data, instructions and intermediate results during processing.

Types covered:
RAM (Main Memory)
Cache Memory

2. RAM (Main Memory)

Purpose

Stores programs and data currently in use by the CPU.

Characteristics

- Large capacity
- Slower than cache memory
- Volatile (data lost when power is off)
- Directly accessible by CPU

Types of RAM

- DRAM (Dynamic RAM)
- SRAM (Static RAM)

3. Cache Memory

Purpose

Small, very fast memory between CPU and RAM that stores frequently used data/instructions to speed up processing.

Characteristics

- Very fast
- Small capacity
- More expensive per bit
- Acts as a buffer between CPU and RAM

Levels of Cache

- L1 Cache – Smallest & fastest, inside CPU
- L2 Cache – Larger, a bit slower
- L3 Cache – Largest, slower than L1 & L2

4. RAM vs Cache Memory

Feature	RAM	Cache Memory
Location	Main memory chips	On or near CPU
Speed	Slower	Much faster
Capacity	Large (GBs)	Small (KBs to MBs)
Cost per bit	Low	High
Volatility	Volatile	Volatile
Access	Directly by CPU	Automatically managed
Purpose	Stores programs & data in use	Speeds up CPU by storing frequently used data/instructions

5. How They Work Together

CPU requests data/instructions.

1st, CPU checks Cache.

If found in Cache (Cache Hit) → served quickly.

If not found (Cache Miss) → data is fetched from RAM, stored in Cache, and then sent to CPU.

This reduces access time and improves overall system performance.

Conclusion

RAM provides large storage for programs and data, while Cache memory provides very fast access to frequently used data. Together, they work in hierarchy to improve CPU performance and overall system efficiency.

Computer Buses

(Data Bus, Address Bus and Control Bus)

1. Data Bus

Carries data between the CPU, memory and I/O devices.

Features

- Bidirectional (data can flow both ways)
- Width determines how much data can be transferred at a time
- Example: 8-bit, 16-bit, 32-bit, 64-bit

Example



Width

Determines performance



2. Address Bus

Carries the addresses of memory locations or I/O devices.

Features

- Unidirectional (from CPU to memory/I/O)
- Determines the maximum memory size that can be addressed
- n address lines can address 2^n locations

Example



Address Lines vs Memory

Address Lines (n)	Max Memory Addressable
10	$2^{10} = 1K$
16	$2^{16} = 64K$
20	$2^{20} = 1M$
32	$2^{32} = 4G$

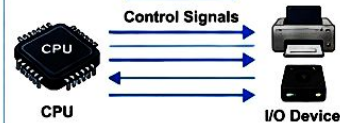
3. Control Bus

Carries control signals to coordinate operations between CPU, memory and I/O devices.

Common Control Signals

- Read (RD) – Read data from memory/I/O
- Write (WR) – Write data to memory/I/O
- Memory Enable (MEM EN)
- I/O Enable (IO EN)
- Interrupt Request (INT)
- Acknowledge (ACK)
- Clock (CLK)
- Reset (RST)

Example



Role

Ensures proper timing and coordination of all operations in the system.

4. How Buses Work Together



CPU initiates an operation.



Address Bus sends the address of the memory location or I/O device.



Control Bus sends control signals (Read/Write, Enable, Clock, etc.).



Data Bus transfers data between CPU, memory and I/O devices.



Operation completed successfully.

Importance of Computer Buses



Enable communication between all components.



Affect system performance and speed.



Proper coordination ensures efficient data transfer.

Conclusion



Data Bus transfers data, Address Bus specifies the location, and Control Bus coordinates the operations. Together, they form the backbone of communication in a computer system.



Computer Components and Their Functions



1. Input Devices



Used to enter data and instructions into the computer.

Examples

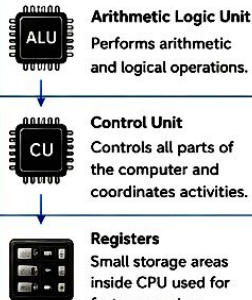
	Keyboard Enters text and numbers.
	Mouse Controls pointer and selects items.
	Scanner Converts images or documents to digital form.
	Microphone Inputs sound or voice.
	Webcam Inputs video and images.

2. Processing Device (CPU)



Performs all calculations, processes data and controls all operations.

Components



3. Memory / Storage Devices



Stores data, instructions and results.

Primary Memory (Main Memory)



Used by CPU directly. Volatile memory.

- RAM (Random Access Memory)
- ROM (Read Only Memory)

Secondary Memory (Auxiliary Storage)



Stores data permanently. Non-volatile memory.

- Hard Disk Drive
- SSD (Solid State Drive)
- USB Flash Drive
- CD / DVD

4. Output Devices



Display or produce the processed information as output.

Examples

	Monitor Displays text, images and videos.
	Printer Produces hard copy of documents.
	Speakers / Headphones Produces sound or audio output.
	Projector Displays large images on screen.

5. Communication Devices



Used to communicate and exchange data between computers.

Examples

	Network Card (NIC) Connects computer to a network.
	Modem Modulates / Demodulates signals for transmission.
	Wi-Fi Adapter Provides wireless network connectivity.
	Bluetooth Device Enables short-range wireless communication.

How They Work Together



Conclusion

All the components of a computer system work together to perform a task efficiently.
Each component has a specific function, and together they make the computer system complete and useful.



Embedded Systems

(Definition, Components, Types, Applications, Advantages)

1. Definition

An Embedded System is a special-purpose computer system designed to perform specific tasks within a larger system.



2. Components



Microcontroller / Microprocessor
(Brain of the system)



Memory (RAM / ROM)
Stores programs and data



Input / Output Interfaces
(Buttons, Sensors, Actuators, Display, etc.)



Power Supply
Provides required electrical energy



Software (Firmware)
Controls hardware and performs tasks

3. Types of Embedded Systems

Stand-alone Embedded System



Works independently to perform specific tasks.
Example: Microwave Oven

Real-time Embedded System



Responds to inputs within a strict time constraint.
Example: Pacemaker

Networked Embedded System



Connected to a network to communicate and share data.
Example: Smart Camera



4. Applications



Consumer Electronics
(TV, Camera, Washing Machine)



Automotive Systems
(ABS, Airbag, Engine Control)



Industrial Automation
(Robotic Arms, PLCs)



Medical Devices
(Heart Monitor, Infusion Pump)



Smart Home Devices
(Smart Thermostat, Lights)



Aerospace and Defense
(Avionics, Navigation Systems)

5. Advantages



High Performance
Designed for specific tasks, so highly efficient.



Reliability
Works reliably for long periods.



Low Power Consumption
Consumes less power, suitable for portable devices.



Cost Effective
Reduces overall system cost.



Compact Size
Small in size and lightweight.



Safety and Security
Can be designed for critical and real-time environments.

6. How Embedded Systems Work

Input from Sensors or Devices



Microcontroller / Microprocessor
Processes the input



Software (Firmware)
Executes instructions and controls operations



Output to Actuators or Display



Performs the Desired Task/Function



Conclusion

Embedded systems are an integral part of modern life. They are specially designed to perform dedicated tasks efficiently, reliably and with low power. They are used in a wide range of applications to make devices smarter and automated.

